Name ____________________

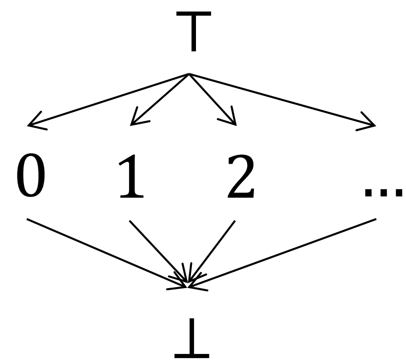
Thursday, November 19

Email ____________________

6.818 Fall 2020 | Miniquiz #26 | 5 Minutes

Assume you want to statically analyze the following MITScript VM bytecode program in order to determine the size of the operand stack at each point in the program:

```
1: if 3
2: load_const 0
3: load_const 1
4: call 3
5: neg
```



Furthermore, assume that stack sizes are represented using the flat lattice of integers as shown on the right.

1. For each type of instruction used in the program above, define a transfer function that computes the stack size right after the instruction is executed as a function of the stack size right before the instruction is executed (s_i):

   $T_{if\ m}(s_i) =$

   $T_{load\_const\ m}(s_i) =$

   $T_{call\ m}(s_i) =$

   $T_{neg}(s_i) =$

2. Define a system of equations that represent the stack sizes s_i after each instruction i in the program above is executed, and then solve the system of equations. Assume that the stack initially contains two elements (i.e., $s_0 = 2$).

   $s_1 =$

   $s_2 =$

   $s_3 =$

   $s_4 =$

   $s_5 =$